Guru Nanak Dev Engineering College, Ludhiana

Department of Computer Science & Engineering

LABORATORY FILE

Artificial Intelligence

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Class : Second Year — Div C2

Subject : Artificial Intelligence

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List of Practicals

Sr. No.	Experiment Title	Date	Sign	Marks
1	Implement a simple reflex agent for the vacuum-cleaner world.			
2	Implement BFS and DFS for water jug problem.			
3	Implement Hill Climbing search algorithm.			
4	Implement A* algorithm for eight puzzle problem.			
5	Develop Tic Tac Toe using Minimax with alpha-beta pruning.			
6	Represent a real-world domain using frames and semantic networks.			
7	Implement Goal Stack Planning for the Blocks World.			
8	Implement Bayes Rule to classify emails as spam or not spam.			
9	Build a small neural network from scratch.			
10	Generate images from text using a pre-trained diffusion/GAN model.			

Experiment No: 1 Mantej Singh Josan — 2435087
Date: 01/01/2025

Implement a simple reflex agent for the vacuum-cleaner world.

Aim

To implement a simple reflex agent that acts based on the current percept only, demonstrated using the vacuum-cleaner world.

Theory

A simple reflex agent selects actions based only on the current percept, ignoring all prior history. It uses condition-action rules of the form *if percept then action*. The vacuum-cleaner world consists of two locations (A and B), each of which can be dirty or clean. The agent perceives its current location and whether it is dirty, then acts accordingly — either cleaning or moving to the other location.

Code

```python
# Vacuum Cleaner World – Simple Reflex Agent

def reflex_agent(location, status):
    if status == "Dirty":
        return "Suck"
    elif location == "A":
        return "MoveRight"
    else:
        return "MoveLeft"

# Test the agent
world = [("A", "Dirty"), ("A", "Clean"), ("B", "Dirty"), ("B", "Clean")]

print("Location | Status | Action")
print("-" * 35)
for loc, stat in world:
    action = reflex_agent(loc, stat)
    print(f"  {loc:<8} | {stat:<6} | {action}")
```

Output

The program was executed and the output is shown on the following page.

Output Screenshot

Conclusion

The simple reflex agent was successfully implemented for the vacuum-cleaner world. The agent correctly determines its action based solely on the current percept without any memory of past states. This demonstrates the basic structure of a condition-action rule-based agent.

Experiment No: 2 Mantej Singh Josan — 2435087
Date:

Implement BFS and DFS for water jug problem.

Aim

Theory

Some theory needed for the topic of practical

Code

```
# Vacuum Cleaner World – Simple Reflex Agent

def yourfunction():
    pass

# Your logic

print("hello world")
print("-" * 35)
```

Output

The program was executed and the output is shown on the following page.

Output Screenshot

Conclusion

What you learnt

Date:

Faculty Signature: _____